
Python for Everybody

Course 2:

Python Data Structure

STRINGS(CHAPTER-6)

- String is a sequence of characters.
- String literals use quotes, either single quotes (') or double quotes (").
- String literals use '+' operator for concatenation of string values.

```
>>> str1 = "Hello"
>>> str2 = 'there'
>>> bob = str1 + str2
>>> print(bob)
Hellothere
>>> str2 = '123'
```

- Every single character in a string is indexed with a specific index, and that mainly starts from zero.



A diagram showing the string 'banana' with each character in a separate box. Below each box is its corresponding index: 'b' at 0, 'a' at 1, 'n' at 2, 'a' at 3, 'n' at 4, and 'a' at 5.

```
>>> fruit = 'banana'
>>> letter = fruit[1]
>>> print(letter)
a
```

- Indexing out of range probably gives an "out of range" error.

- **String conversion:**
- Numeric string can be converted into integer by using `int()` or `float()`.
- **Len function:**
- `len` is an built-in python string function ,that gives the length of the string.
- **Loop through string:**
- Looping through string is possible by using definite and indefinite loop.
- But using for loop is much more elegant than that of using while loop since the number of line of code used in for loop is less than that of while loop.

```
fruit = 'banana'
for letter in fruit :
    print(letter)
```

```
index = 0
while index < len(fruit) :
    letter = fruit[index]
    print(letter)
    index = index + 1
```

- **Slicing string:**
- String slicing in Python is a powerful and flexible method for extracting portions of a string, known as substrings, by specifying a range of indices.
- It operates on the principle that strings are ordered sequences of characters, each with an associated index starting from 0.

- The second number is one of beyond the end of the slice”up to included but not included”.

```
>>> s = 'Monty Python'
>>> print(s[0:4])
Mont
>>> print(s[6:7])
P
```

- If the second number is beyond the range then we stop in the end of the loop.
- If we leave off the first number or the last number of the slice, it is assumed that the string starts from the beginning and ends at the end.

```
>>> s = 'Monty Python'
>>> print(s[:2])
Mo
>>> print(s[8:])
thon
```

- In:
- In keyword in string is used for checking if two strings are equal or not.
- And gives us output in terms of either true or false.
- String comparison:
- By using comparison operator we can compare two string.
- Lowercase letters are always > uppercase letter ,example 'a'>'A'

```
if word == 'banana':  
    print('All right, bananas.')  
  
if word < 'banana':  
    print('Your word,' + word + ', comes before banana.')  
elif word > 'banana':  
    print('Your word,' + word + ', comes after banana.')  
else:  
    print('All right, bananas.')
```

- String library:
- Python has number of string function within string library.
- These function are already built into each string - we can use them by invoking them , by appending the function to the variable.
- These string function do not modify the original string,instead they create a copy and returns the altered string.

- Some of the common functions:

```
str.capitalize()  
str.center(width[, fillchar])  
str.endswith(suffix[, start[, end]])  
str.find(sub[, start[, end]])  
str.lstrip([chars])
```

```
str.replace(old, new[, count])  
str.lower()  
str.rstrip([chars])  
str.strip([chars])  
str.upper()
```